

QEM Zero–Twelve Dimensional Dual-State Cosmological Framework

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Abstract

This paper proposes the QEM Zero–Twelve Dimensional Dual-State Cosmological Framework, a model in which the zero-dimensional origin and the twelve-dimensional maximal structure are not independent scales but two faces of a single negative-complex dimensional identity. Within this framework, the smallest observable scale and the largest cosmic scale converge to the same structural state through a finite dimensional ladder, forming a self-observation universe driven by an active–passive hierarchy of observers. The model provides an intuitive geometric picture for how observer-dependent collapse, multi-layer causality, and cosmic-scale coherence can emerge from a single dual-state rule. It is intended as a phenomenological, structure-first proposal rather than a finished physical theory.

Keywords

Zero–Twelve Dimension; Dual-State Universe; Negative-Complex Dimension; AP-Hierarchy; Observer Flip; Self-Observation Universe; QEM Cosmology

1. Introduction

Modern physics treats the smallest and largest scales as conceptually distinct. Quantum field theory describes microscopic particles, while cosmology describes the evolution of the universe at the largest scales. Between these regimes, a variety of effective theories are used. However, there is still no single structural rule that explains why the universe appears to be both discrete and continuous, both local and nonlocal, and both observer-dependent and observer-independent.

The QEM program (Quantum Emergent Mesh) approaches this problem from a structure-first perspective. Instead of starting from known equations and searching for unification, it starts from simple combinatorial and geometric constraints and asks: what kind of universe necessarily emerges when these constraints are applied at all scales? In previous work, QEM-8³ and OCN-3D described a closed network of oppositional states built from eight basic configurations and three-dimensional closure rules. In this paper we extend the program to the cosmological scale by introducing a dual-state relation between zero and twelve

dimensions.

2. Dimensional Ladder and the Zero–Twelve Closure

We consider a finite ladder of effective dimensions labelled 0, 1, 2, ..., 12. Each rung of the ladder represents a distinct level of structural freedom: for example, 0-D for point-like closure, 1-D for linear propagation, 2-D for surface tilings, and so on. The crucial postulate is that this ladder does not extend indefinitely. Instead, at the twelfth rung the structure re-enters the zero-dimensional state, forming a closed loop.

Formally, we introduce a mapping F such that $F(0) = 12$ and $F(12) = 0$, while intermediate dimensions map to themselves under a single step of the ladder. The pair (0, 12) is thus treated as a single dual-state element. From the viewpoint of an internal observer living on any finite rung 1–11, zero and twelve can be distinguished; from the viewpoint of the global structure, they are facets of one object.

3. Negative-Complex Dimensional Interpretation

To encode the dual-state nature of the zero–twelve pair, we introduce an effective parameter D defined as a negative-complex quantity:

$$D = -(12 + i \cdot \Omega),$$

where Ω represents the internal rotational phase that couples microscopic and macroscopic scales. The minus sign indicates that the dual-state is not simply an additional positive dimension, but an inversion of all intermediate ladders. The imaginary component embodies the cyclic, phase-like behaviour of the system: as Ω winds through a full cycle, the structure interpolates between appearances that resemble a zero-dimensional point and a twelve-dimensional maximal network.

We do not claim that D is a conventional dimension in the mathematical sense. Rather, it is a bookkeeping device that captures the idea that the smallest and largest structures are mathematically identical after an appropriate inversion and rotation in an abstract space.

4. Active–Passive Hierarchy and Observer Flip

A central theme of the QEM program is the Active–Passive (AP) hierarchy. Instead of assuming a single, absolute observer, we treat observation itself as a layered process. At each

dimensional rung there are subsystems that can play an 'active' role (initiating measurements, setting boundary conditions, or triggering updates) and subsystems that play a 'passive' role (being measured, responding to imposed constraints).

Observation is modelled as a flip between these roles. When a subsystem becomes active with respect to another, it defines a local classical frame and induces an effective collapse of possible configurations into a realized pattern. However, from the viewpoint of a larger layer in the hierarchy, both subsystems may appear passive components of a higher-order process. In this sense, every observer is simultaneously an observed object at some higher level.

The zero–twelve dual-state provides a natural top and bottom for this hierarchy. At the zero-level, observation is represented as the minimal possible disturbance: a single flip in the smallest available unit. At the twelve-level, observation corresponds to a collective, cosmic-scale coordination of many such flips. Because 0 and 12 are dual, these two descriptions are mathematically equivalent.

5. Self-Observation Universe

If the dimensional ladder closes between zero and twelve, then there is no need to posit an external, transcendent observer outside the universe. Instead, the universe can be viewed as a self-observing structure. Its smallest units act as microscopic observers for local events, while its global configuration acts as a macroscopic observer for large-scale patterns. The AP hierarchy mediates between these levels.

This picture resonates with various ideas in quantum foundations, where measurement is treated as an interaction between subsystems rather than an operation by an external agent. In the QEM zero–twelve framework, the ladder structure ensures that such interactions can propagate consistently across scales without leading to infinite regress. The top of the ladder is already identified with the bottom.

6. Simple Structural Model

To make the discussion more concrete, we outline a minimal structural model. Consider a network of nodes arranged such that each node carries a discrete state drawn from a finite alphabet. Local update rules implement opposition and closure constraints similar to those used in QEM-8³ and OCN-3D. The network is organized into layers corresponding to effective dimensions: neighbourhood size, connectivity, and allowed cycles increase

systematically from layer 0 to layer 12.

At layer 0, a node only interacts with itself; updates are trivial but define the basic unit of disturbance. At intermediate layers, nodes interact with larger neighbourhoods, producing extended patterns that can be interpreted as quasi-particles, fields, or geometric configurations. At layer 12, the network is maximally connected under the closure rules.

The dual-state postulate is implemented by requiring that a full cycle of updates at layer 12 reproduces, in compressed form, the state of an effective layer-0 node. Conversely, an update at layer 0 is defined to encode the summary of many microscopic events distributed across higher layers. In this way, local and global descriptions are constrained to agree.

7. Conceptual Consequences

Several conceptual consequences follow from the zero–twelve dual-state framework:

(1) Scale Identification: The smallest and largest scales are not fundamentally different categories but different projections of the same underlying structure.

(2) Finite Hierarchy: The dimensional ladder terminates after a finite number of steps. There is no need to assume infinite regress in either direction; beyond twelve, one has already returned to zero.

(3) Internal Observer Principle: Every observer is a subsystem within the universe. Self-observation is possible because the ladder closes, allowing information to cycle between local and global states.

(4) Negative-Complex Encoding: The behaviour of the dual-state can be represented mathematically by an effective negative-complex parameter. This suggests that certain cosmological phenomena (such as apparent acceleration or large-scale coherence) might be re-interpreted as phase effects of this parameter.

(5) Bridge to AP-Hierarchy: The AP hierarchy of active and passive roles finds a natural geometric basis in the ladder.

Upward transitions correspond to a subsystem becoming part of a larger observer, while downward transitions correspond to the coarse-graining of many events into an effective single flip.

8. Outlook

The present paper is intentionally modest in its mathematical detail. Its main aim is to set out a clear structural proposal: that zero and twelve dimensions should be regarded as a single dual-state object, forming a closed ladder of effective dimensions in which the universe observes itself through an active–passive hierarchy.

Future work will focus on three directions. First, we plan to develop explicit lattice models that instantiate the dual-state rule and allow numerical exploration of its emergent behaviour. Second, we aim to connect the framework with existing approaches in quantum gravity and cosmology, looking for potential observational signatures. Third, we will study the implications for information theory and consciousness research, where self-observation and multi-scale feedback already play a central role.

The QEM Zero–Twelve Dimensional Dual-State Cosmological Framework is not proposed as a final answer, but as a compact, testable way to encode several intuitions: that the universe may be finite yet unbounded in its structural hierarchy, that observers and observed systems are different aspects of the same network, and that the smallest and largest scales may in fact be one and the same when seen from the right structural angle.